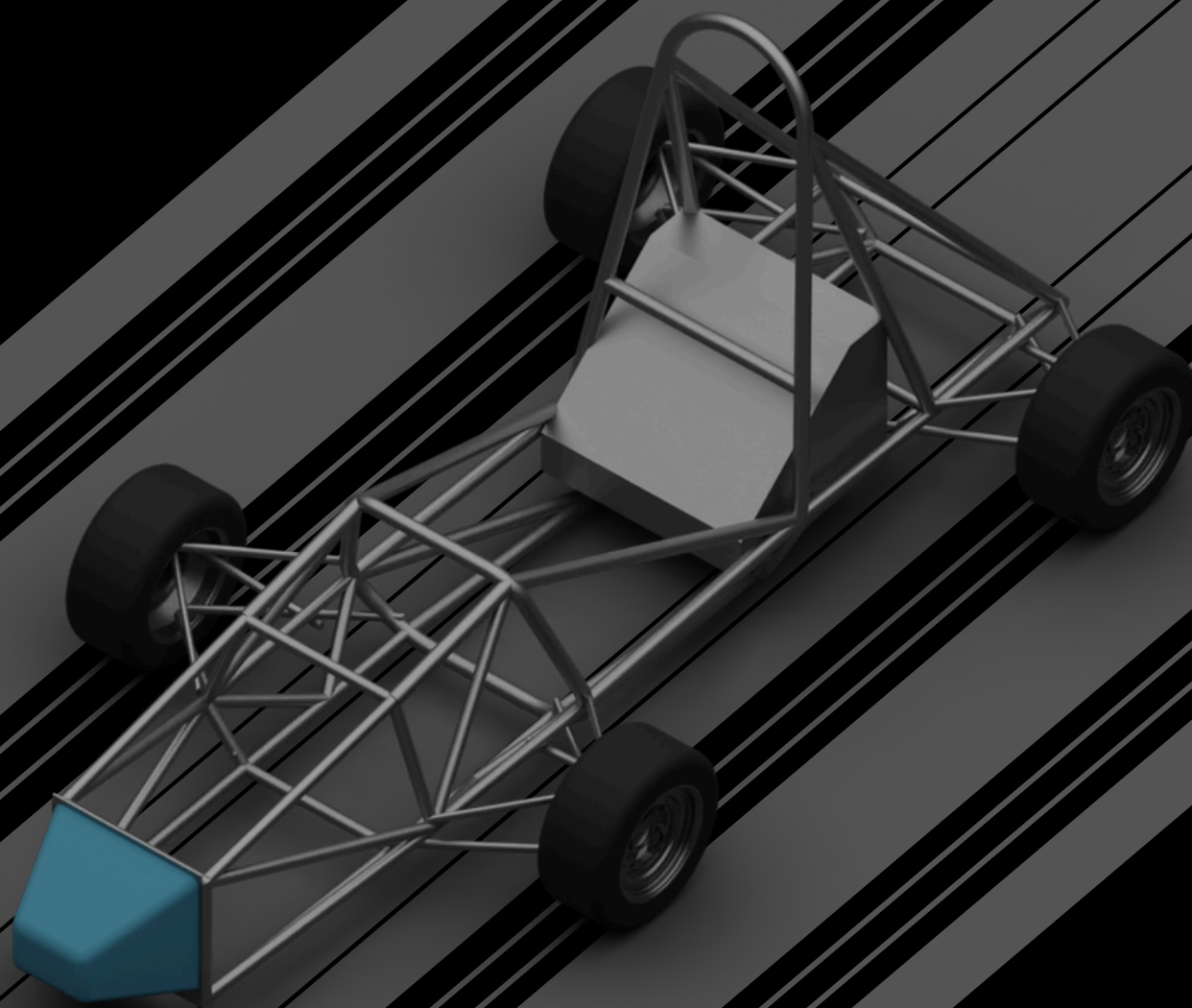




NEWSLETTER

SEPTEMBER 2024



CONTROL ARMS

YOU MAY HAVE HEARD
OF DOUBLE WISHBONE.
BUT WHY?

MANAGING THE BATTERY

LEARN ABOUT THE
IMPORTANCE OF A BMS

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WHAT IS AUS RACING

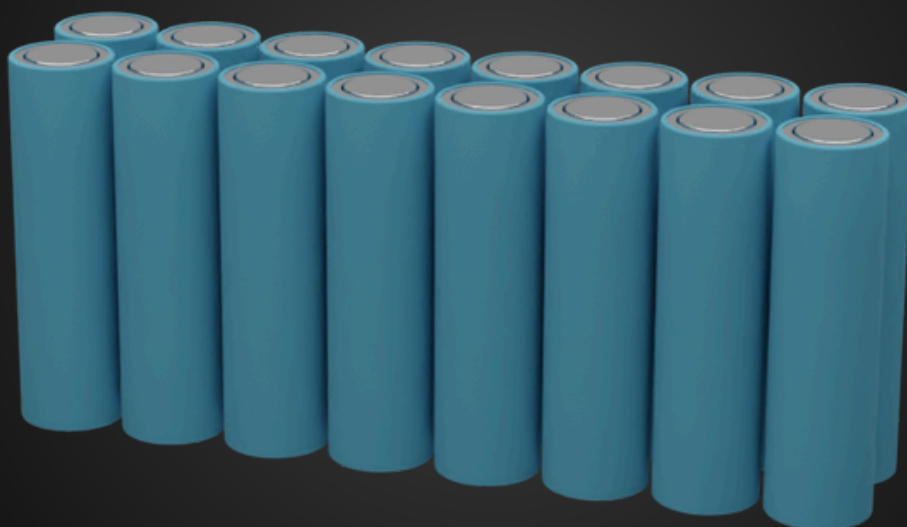
AUS Racing is the Formula Student team from the American University of Sharjah. Formula Student is an international engineering design competition that graciously provides a platform to aspiring engineers to build an open-wheel, single seater racecar and simulate all the different nuances that culminate to account for its structural integrity, performance, economic feasibility and industrial standards.

Students compete in different static and dynamic events to assess the car, thus testing the engineering design, management skills and the overall team spirit of the contesting teams. It is an esteemed competition that unlocks new horizons for AUSers

to venture into as we progress into a new era of technological competence and innovation.

Here at AUS Racing, we hope to open up opportunities for students for the foreseeable future to take part in a project that will teach them engineering, management and budgeting. In essence, real world engineering.

MOUNTING THE CELLS



Mounting the cells securely is crucial to the entire system. A cell coming loose can lead to catastrophic failure.

Furthermore, mechanical and electrical integrity go hand in hand while designing cell housings. A bad mechanical design can lead to sub-optimal electrical results.

The cell housing also needs to be insulating and fire resistant while being able to conduct electricity through the electrical connections.

Modularity is also crucial for making changes easily as our designs will require a lot of iteration under a tight budget.

MANAGING THE BATTERY



The Battery Management System, also known as BMS, is the most crucial system in the entire lithium-ion power delivery system.

The BMS has a variety of tasks, both while charging and while supplying power to the motor and other components.

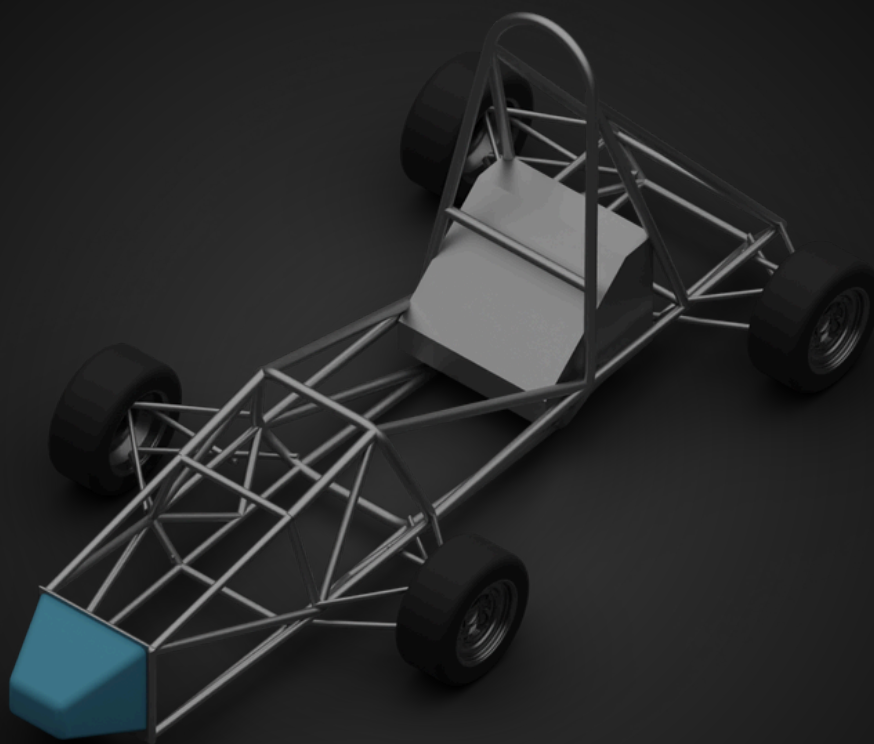
While charging, the BMS needs to ensure that all the cells are

being charged at the same rate without overheating and preventing excessive current.

While supplying power i.e. discharging, the BMS manages the temperature and protects the cells from exceeding their maximum voltage.

BMS is also the main point of communication between the battery and the rest of the system

GUIDING THE WHEELS



The control arms guide the wheels of a car to ensure it only moves along one dimension effectively as needed.

Our car features a double wishbone style suspension, similar to F1 suspension. This type of suspension allows the most freedom for fine tuning post manufacturing, making it the ideal choice.

The suspension, as the name

suggests, has two control arms shaped like wishbones to support the upright with wheels.

The control arms do not need to be too heavy-duty as the kind of stresses experienced are mostly axial.

Designing the control arms are the first step towards designing the optimal uprights, and hence, an optimal outer suspension system.

OUR MEETING WITH ANSYS



Our drivetrain is comprised of a simple system, however, it is a very intricate system. It involves extensive research and development to design an effective drivetrain with minimum energy losses and maximizing reliability.

Our drivetrain comprises of a simple chain drive to ensure the the speed and torque at the wheels is exactly what we need. A transmission is not required as

the electric motor always delivers the ideal power.

Getting the torque from the chain to the wheel is a challenge in itself. Unfortunately, the wheel is never at the same level relative to the chassis because of the suspension system.

To circumvent this, tripod joints are used. They are a special kind of joint that allow up to 12° of motion in 2 axes.

OUR VISIT TO AUTOMECH



The AUS Racing team recently visited Automech Group's machine shop, marking a significant step in our Formula Student journey. As our build partner, Automech will provide us with the infrastructure and technical expertise needed to bring our design from concept to reality for the upcoming FSAE competition.

During the visit, the team explored the workshop and learned more about the advanced machinery and fabrication processes that will play a vital role in constructing our car.

Automech's support ensures we have access to the highest-quality resources, enabling us to focus on precision and performance in our build.

OUR GREEN INITIATIVE

We are excited to officially inaugurate the Department of Sustainability at AUS Racing. As a Formula Student team, we understand the significance of sustainability in engineering and aim to incorporate responsible practices in every aspect of our project.

The new department will focus on integrating sustainable strategies into our design and manufacturing processes, such as optimizing material selection, minimizing waste, and reducing the carbon footprint of our car. By adopting these principles from the early design stages, we strive to build an electric race-car that is also environmentally responsible.

Additionally, the Department of Sustainability will work to promote awareness within the team and our wider community, highlighting how engineering can align with environmental values. Through team initiatives and outreach, they will lead efforts to champion sustainability in our project and inspire others.

We look forward to seeing how this department will help AUS Racing set new standards, combining technical excellence with a commitment to a greener future.



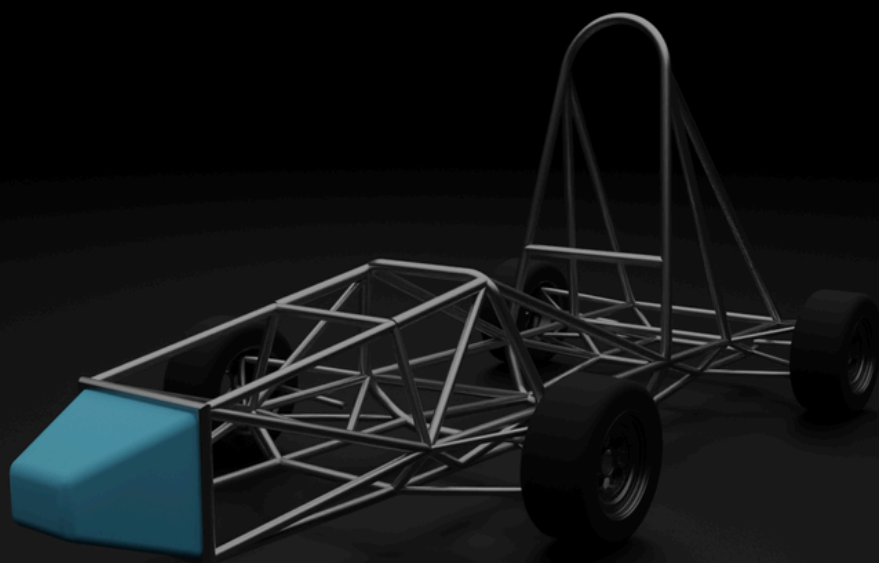
HOW YOU CAN SPONSOR US

What if you were at the forefront of engineering innovation?

What if your company was responsible for bringing our aspiring engineers' ideas to life?

What if your business could get its message across the world through Formula Student?

If your answer to any of these questions is yes, click [here](#) and check out what we have in store for our generous sponsors and learn how to become one yourself!



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